Filbin 2018 pediatric H3-K27M glioma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant structure is sample-mixed AND continuous. iLISI 1.00→3.81 of 6 samples after Harmony(θ=10); one connected component on the independent Harmony-PCA graph (no islands removed). Gast is not proliferation-confounded (prolif–Gast ρ=+0.09).

Gate 2 — module geography → GO (non-branching)

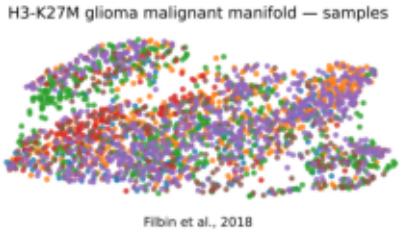
Non-branching embryonic axis. Gast/pSom co-localise (joint enrichment 1.54; spatial r=+0.61) as one developmental origin; Crest is an inverse terminal gradient (pSom&Crest joint 0.22; Crest-high decile pseudotime 0.99). No Gast/pSom hub diverging into two arms — GO (non-branching).

Gate 3 — cross-repression → NA

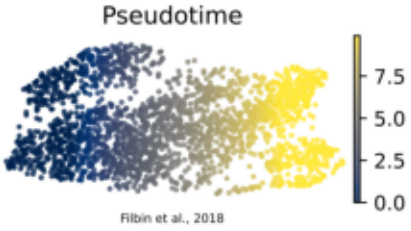
Not applicable — no bifurcation. A single principal curve (2 tips, 0 forks) fits the manifold, so there is no branch point at which to test mutual Gast↔pSom cross-repression.

STEP FIGURES

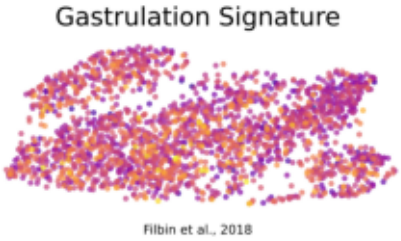
Manifold — samples



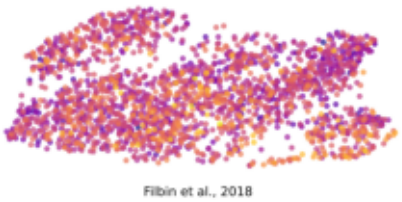
Pseudotime



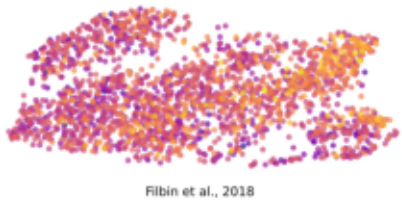
Gastrulation



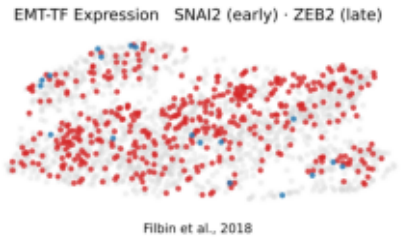
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric H3-K27M mutant glioma (Filbin et al., 2018), SmartSeq2; 2,259 malignant cells across 6 samples (all kept, ≥100 floor). One continuous non-branching manifold.

Verdict.
Pattern reproduces (non-branching). One continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high origin; Crest a terminal inverse gradient. G1 GO · G2 GO · G3 NA — Linear-pSom/Crest.

Parameter choices

Min-cell threshold	≥100 floor → all 6 samples kept (2,259 cells); 141–667 smooth, no cliff	**Integration**	Harmony on sample, θ=10, n_pcs=20; patient batch 0.96, plateaus by θ=5
Geometry	scFates curve, Nodes=20 — non-branching (2 tips, 0 forks; self-approach 2.97)	**Root node**	Gast/pSom-high, Crest-low tip 2 — developmental origin; non-cycling
Islands removed	0 — single connected component (UMAP + Harmony-PCA graph)	**EMT-TFs shown**	SNAI2 (early) · ZEB2 (late) — best-demarcating pair; ZEB1/ZEB2/PRRX1 broad